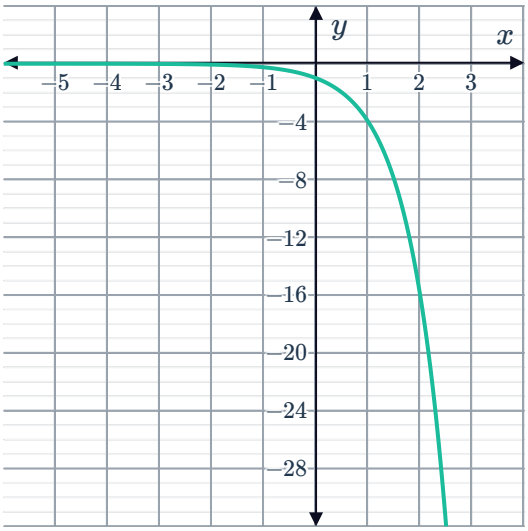
Exponential and quadratic functions

1 Consider the following two functions:

- Function A



- Function B

$$y = -\frac{4\,(x-3)\,(x-7)}{21}$$

a Determine which function has the greater function value at the y-intercept.

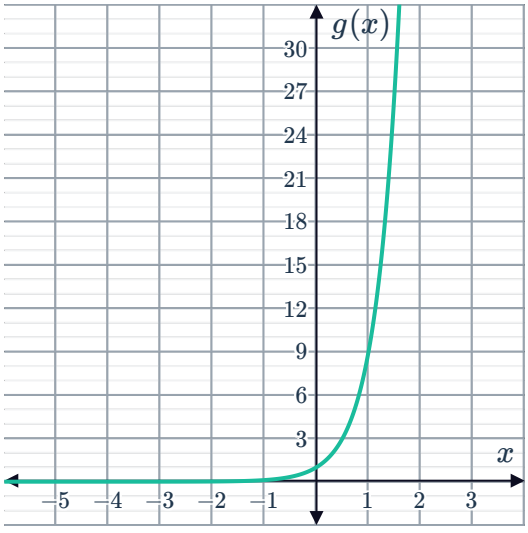
b Determine if each function is increasing or decreasing on the domain $0 < x < 5$.

2 Consider the two functions $f\,(x)$and $g\,(x)$:

x	-2	-1	0	1	2
$f(x)$	36	9	0	9	36

a Determine which function has a minimum value.

b Determine which function has a horizontal asymptote.

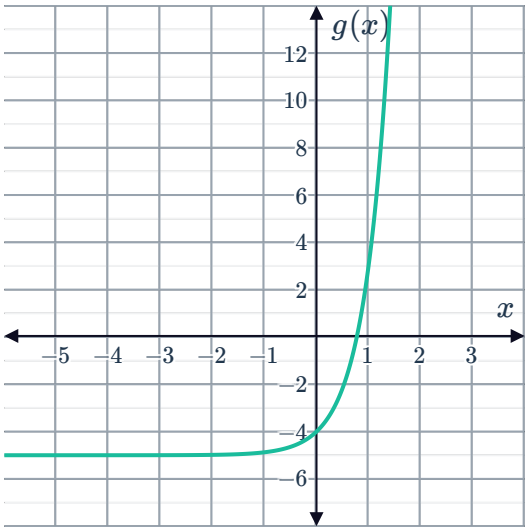


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3 Consider the function $f(x) = 8x^2 - 5$ and the function $g(x)$ whose graph is shown:

a Determine which function is increasing for $x < 0$.

b Determine which function is increasing for $x > 0$.



4 The exponential function P is given by $y = 4^x + 4$ and the parabola Q is given by $y = \dfrac{(x-6)(x-2)}{4}$.

a Graph the exponential function P and the parabola Q on the same coordinate plane.

b Determine which function has the greater function value at $x = 4$.

c Determine if each function is increasing or decreasing on the domain $1 < x < 3$.

5 The table of values for the functions P and Q are provided:
Function P:

x	-7	-6	-5	-4	-3	-2
y	-7	-4	-3	-4	-7	-12

Function Q:

x	-10	-5	-1	1	5	10
y	$1\,073\,741\,824$	$32\,768$	8	$\frac{1}{8}$	$\frac{1}{32\,768}$	$\frac{1}{1\,073\,741\,824}$

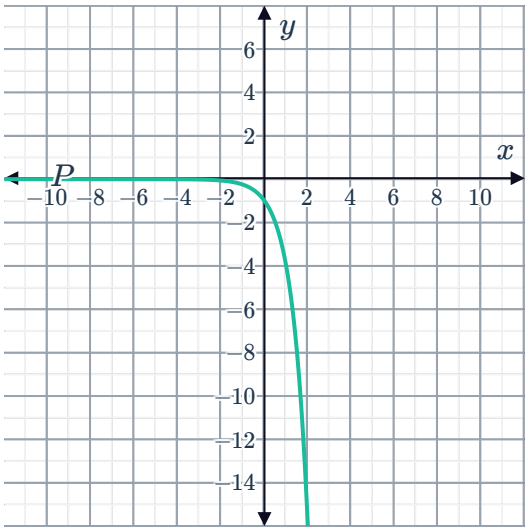
a Determine what type of functions Function P and Function Q are.

b Graph the functions on the same coordinate plane.

c Determine how many times functions P and Q intersect.

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6 The graph of the exponential function P, given by $y = -4^x$ is shown:
Function Q is given by the equation $y = -\dfrac{4\,(x-3)\,(x-7)}{21}$.



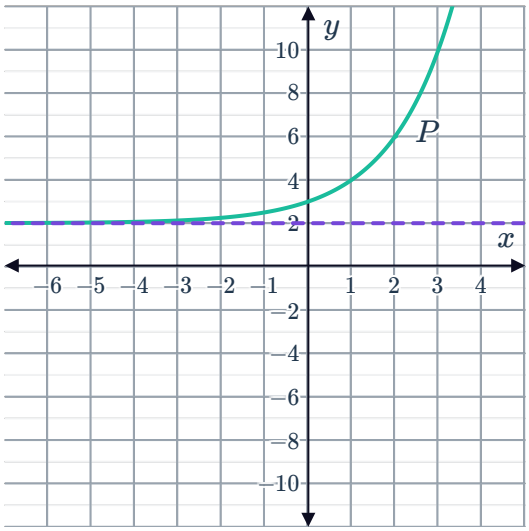
a Sketch the graph of function Q on the same coordinate plane as function P.

b How many times do P and Q intersect?

c Which function has the greater function value at the y-intercept?

d Describe the behavior of functions P and Q on the domain $0 < x < 5$.

7 Consider the graph of the exponential function P and the table of values for parabola Q:
Function $P: y = 2^x + 2$



Parabola $Q: y = -3\,(x+5)\,(x+2) + 2$

x	-5	-4	-3	-2	-1
y	2	8	8	2	-10

a Sketch the graphs of P and Q on the same coordinate plane.

b Describe the behavior of function P and function Q as x approaches infinity.

c Using the graph, determine how many solutions there are to the equation $2^x + 2 + 3\,(x+5)\,(x+2) - 2 = 0$.

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8 Consider the functions $f(x) = 2^x$ and $g(x) = 2x^2$ for $x \geq 0$.

a Complete the table of values for each function:

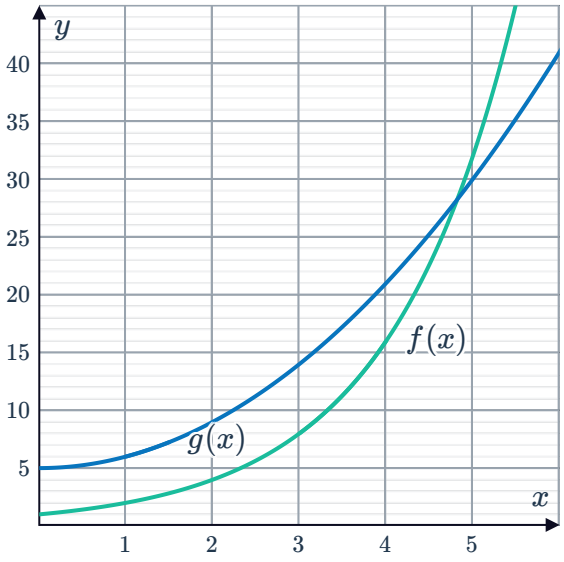
x	$f(x)$	Increase in $f(x)$	$g(x)$	Increase in $g(x)$
0		–		–
1		1		2
2		2		6
3				
4				
5				
6				
7				

b Describe the behavior of the two functions.

9 Kate considers the given graphs of $f(x)$ and $g(x)$ and performs the following calculations:

- Calculation 1: $f(11) - f(10) = 1024$
- Calculation 2: $f(10) - g(10) = 919$
- Calculation 3: $f(11) - g(11) = 1922$
- Calculation 4: $g(11) - g(10) = 21$

She claims that for large values of x, as x increases, the exponential function, $f(x)$, increases more rapidly than the quadratic function, $g(x)$.
Explain how the following pairs of calculations support the claim:



a Calculations 1 and 4.

b Calculations 2 and 3.

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Applications

10 Two mining corporations model the total amount mined, in metric tons, at the end of the nth week of operations. The models are given by:

- Mint Corporation: $M = 100\,(1.15)^{n-1}$
- Crest Corporation: $C = 100n^2$

a Complete the following table of values for the total amount mined by each corporation. Round your answers to the nearest whole number.

n	1	5	10	15
M	100			
C	100			

b Sketch the two functions on the same coordinate plane over the first 65 weeks of operations.

c If mining operations for both companies were to only last at most a year, which company will have mined the most minerals in that time?

d Find the point(s) of intersection of the two graphs and interpret it in this context.

e At the point of intersection the total quantity of minerals remaining in both mines is equal. If both mining companies continue to operate in the same way indefinitely, which company will exhaust their mine first?

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11 To accommodate for its expanding population, a country creates a new city and immediately relocates $400\,000$ of its citizens there. The city's land is allocated such that it can immediately produce enough food for $600\,000$ people in the first year. The table shows the functions that can be used to predict the city's population (P) and the number of people who can be fed (Q), after t years:

Population after t years	$P(t) = 400\,000\,(1.013)^t$
Population that can be fed after t years	$Q(t) = 600\,000\left(\dfrac{t^2}{20\,000} + 1\right)$

a Evaluate $Q(10) - P(10)$ to the nearest whole number.

b Will there be a surplus or shortage of food for the population in 10 years time?

c Evaluate $Q(100) - P(100)$ to the nearest whole number.

d Will there be a surplus or shortage of food for the population in 100 years time?

e Sketch the two functions on the same coordinate plane over the first 150 years.

f Find the point in time when the models predict the city will have just enough food supplies to support its population. Round your answer to two decimal places.

g After the time found in part (f), will the city have enough food to support its population ever again? Explain your answer.

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